

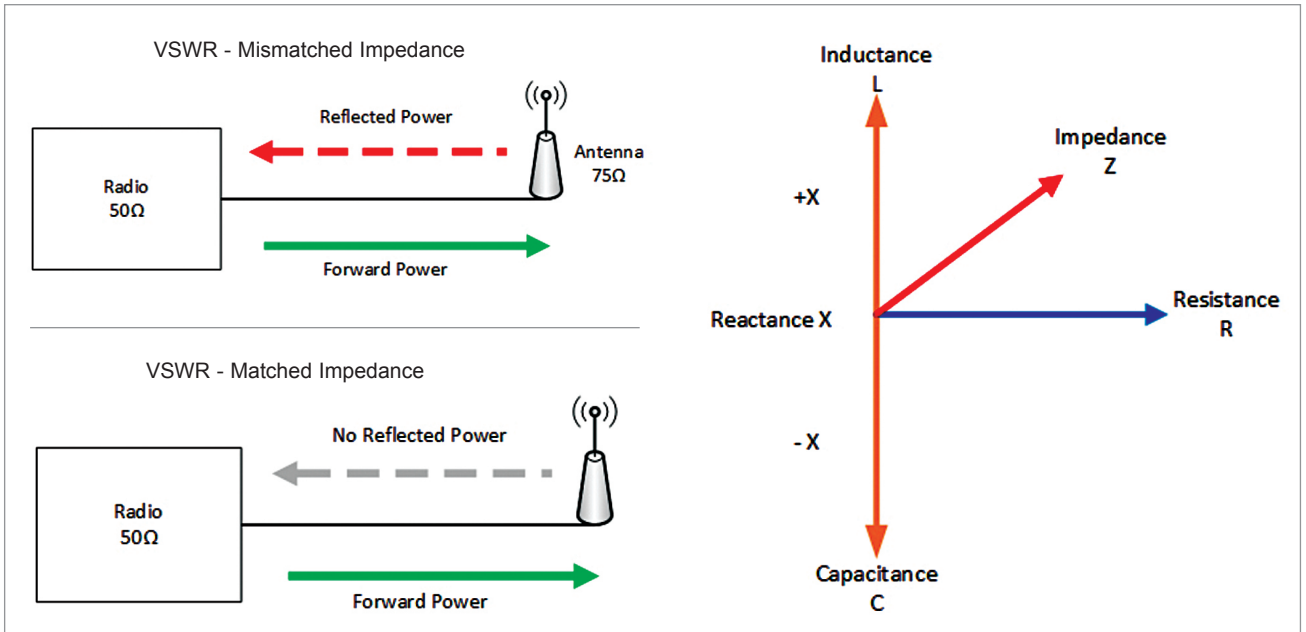


Fig. 2: Mismatched and matched impedance conditions

where,

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Z_L and Z_0 are the load and characteristics impedance of the transmission line.

When there is a proper impedance match between the source and the load, there is no reflection and hence $\Gamma = 0$, and thereby VSWR is the positive and real number with minimum and maximum value of 1 and infinity, respectively. Hence for a perfect tuned antenna, the VSWR should be 1 to indicate that no power is reflected back to the source.

More the VSWR, higher will be the loss and lesser the power transmission efficiency. For the good real-time RF components, VSWR lower than 1.5:1 has to be considered for the typical impedance value of transmission line, which is from 50Ω to 75Ω. The return loss, reflection coefficient, VSWR, and percentage of power reflected are related as shown in Table 1.

From the table we can understand that when the return loss value < 10dB, the antenna will be at VSWR

VSWR	Reflection Coefficient	Return Loss (dB)	% of Reflected Power	Meaning
Infinite	1	0	100	No power transmitted and all reflected (short or open circuit)
17	0.84	1	80	80% reflection and 20% transmission
6	0.44	3	50	50% reflection and 50% transmission A good rule of thumb: 3:1 = 6dB
2	0.32	10	10	10% reflection and 90% transmission A good rule of thumb: 1.5:1 = 14dB
1	0	Infinite	0	0% reflection and 100% transmission a perfect match

VSWR	Return Loss (RL)
Ratio of voltage applied to voltage reflected	Portion of a signal reflected due to discontinuity in line to indicate the loss
Preferred in the connector industry	Preferred in the cable industry
It is a linear measurement	It is a logarithmic measurement
Useful when displaying larger reflections	Useful when displaying very small reflections
$VSWR = (1 + 10^{RL/20}) / (10^{RL/20} - 1)$	$Return Loss = 20 * \log_{10} (VSWR + 1 / VSWR - 1)$

> 2, indicating the system will reflect > 30% power and the performance of antenna will be < 70%. For VSWR = 2, reflection coefficient will be = 1/3 and $20\log(1/3) = -9.54\text{dB}$.

Therefore, VSWR of less than 1.5:1 is considered ideal in RF field, and VSWR of 2:1 is marginally acceptable for more critical loss system for low-power applications. **EFY**